DATATHON 2026: Financial analysis report of the AI Dream team

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Abstract

We analyze a Vietnamese fashion e-commerce dataset (2012 to 2022) spanning 121,930 customers and 646,945 orders across 15 relational tables. Using RFM, based on customer segmentation, and cohort retention analysis, we identify a severe risk of revenue concentration risk, the Champions segment ($\approx 20\% of customers$) accounts for 64.1% of total revenue, alongside a critical onboarding failure with an average retention in $month - 1$ of only 3.34%. For sales forecasting, we develop a tree-based ensemble: $LightGBM + XGBoost + SeasonalNaive$ with walk-forward cross-validation over a 548-day horizon. Our analysis yields five actionable business recommendations targeting an estimated 416M VND in recoverable revenue.

1 Analyzing business data.

The analysis focuses on the central questions: "Who are the valuable customers? Who is leaving? And what actions should the business take?" We apply the RFM (Recognance, Frequency, Monetary) method with quintile scoring (1 to 5) in all 121,930 customers, classifying them into 9 segments. The data is integrated from three sources: dim_customers_rfm.csv (RFM scores + demographics), geography.csv (geographical regions, joined via zip), and fact_orders_enriched.csv (purchasing behavior by category, joined via customer_id). Monetary is calculated from payment_value in payments.csv, excluding canceled orders.

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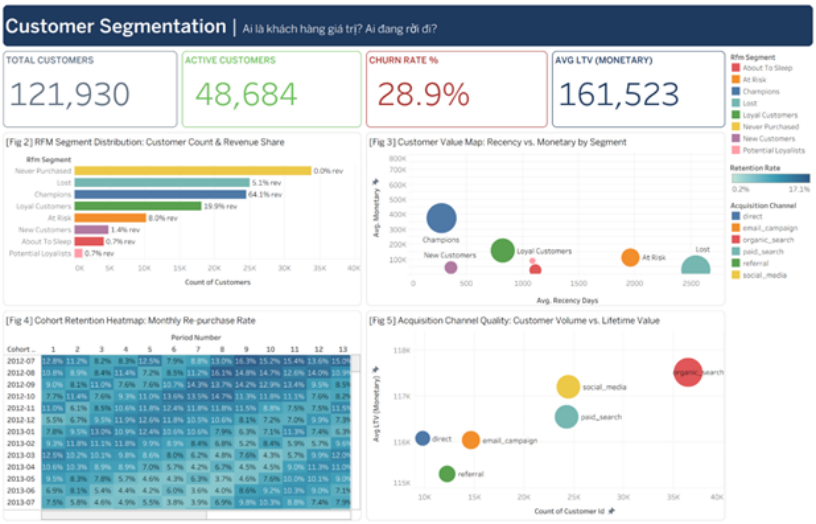
1.1 Descriptive — Current segmentation landscape



Customer and revenue distribution. The total customer base reaches 121,930, of which only 48,684 (39.9%) are classified as active. RFM analysis reveals a highly concentrated revenue distribution: the Champions segment accounts for only about 20% of total customers but contributes 64.1% of total revenue. In contrast, Loyal Customers ($\approx 15\%$) contribute 19.9%, while the remaining five segments (At Risk, New Customers, About To Sleep, Potential Loyalists, Never Purchased) together account for only 15.9% of revenue despite representing $\approx 65\%$ of the customer base. Notably, 27.7% of customers (Never Purchased) have not made any transactions despite registering accounts, this is a segment where acquisition costs have already been incurred but no revenue has been generated.

1.2 Diagnostic — Why does this disparity exist?

Behavior by recency and monetary. The Customer Value Map (Fig 3) clearly shows four behavioral quadrants. Champions are concentrated in the upper-left corner (low recency $\approx$ 200 days, high monetary ($\approx$ 350K VND), reflecting customers who purchased recently and spend heavily. In contrast, the Lost cluster appears in the lower-right corner (recency $\approx 2,500$ days, monetary $\approx$ 20K VND), indicating they have not made purchases for years. The At Risk segment lies in the middle, with recency $\approx 1,000$, days, these customers used to purchase but are gradually disengaging.

The Potential Loyalists paradox. Analysis of AOV per Segment (Fig 2) reveals an important anomaly: Potential Loyalists have the highest AOV ($\approx 55,000$ VND per order), even exceeding Champions ($\approx 25,000$ VND). Combined with the segment definition ($R \geq 3, F \geq 1, M \geq 3$), this group consists of "big spenders" who purchase only 1 or 2 times with high order value. This indicates that the issue is not AOV but purchase frequency, the correct strategy is to increase frequency, not to upsell order value.

Purchasing behavior by category. Combining data from dim_customers_rfm with fact_orders_enriched (joined via customer_id) enables analysis of category preferences by segment. Fig 3 shows that Champions are heavily concentrated in Streetwear. This is an insight that cannot be derived from a single dataset alone and has direct implications for product recommendation strategy.

1.3 Predictive —What will happen?

Cohort-based retention. The Cohort Retention heatmap (Fig 4) reveals two clear patterns. First, retention in the very first month (Period 1) is already low, ranging from 5% to 13% depending on the cohort, indicating that most customers do not return immediately after their first purchase. Second, from Period 2 onward, retention does not recover but remains consistently low, fluctuating between 3 to 16% over the 24 month observation period. This suggests no improvement over time. We can conclude that the issue lies not only in onboarding but also in long-term retention: once customers

fail to return in the first month, the probability of them returning in subsequent months does not increase significantly.

Onboarding quality over time. Fig 5 analyzes Period 1 (the month immediately following the first purchase) across cohorts. The average line is 3.34%, meaning that only slightly more than 3 out of 100 new customers return to make a purchase in the following month. Early cohorts (2012 and 2013) fluctuate above the average, while many later cohorts fall below this threshold, suggesting that new customer retention quality has not improved over time. This represents the most critical leakage point in the customer lifecycle. For every 100 new customers, the business retains only ≈ 3 in the first month, with the majority leaving before having the chance to become Loyal Customers or Champions.

Revenue at risk forecast. Based on the total monetary contribution of the At Risk (8.0%), Lost (5.1%), and About To Sleep (0.7%) segments, approximately $\approx 13.8\%$ of revenue is currently at risk. Without intervention, a large portion of the At Risk group is expected to transition into the Lost segment within the next 6 to 12 months.

1.4 Prescriptive — What should we do?

Optimize acquisition channels. Acquisition Channel Quality (Fig 4) indicates that organic_search is the most effective channel: it has both the largest volume ($\approx 40,000$ customers) and the highest Avg LTV ($\approx 117,500$ VND). Additional analysis from the Channel × Segment heatmap confirms that organic_search has a higher proportion of Champions compared to other channels, while email_campaign and social_media show relatively higher distribution in lower-value segments. Recommendation: increase investment in SEO/organic_search, while reviewing the targeting strategy of email_campaign.

Action matrix by segment:

Segment	Scale	Revenue Share	priority action	Target KPIs
Champions	≈20% Cus	64.1%	VIP program, early access, exclusive discount	Decrease churn rate to <5%/year
Loyal Customers	≈15% Cus	19.9%	Loyalty rewards, upsell Streetwear	Increase frequency +20%
Potential Loyalists	≈5% Cus	0.7%	Re-engagement campaign, tăng tn sut	Increase F score from 1 to ≥3
At Risk	≈8% Cus	8.0%	Win-back: discount 15% + free shipping	Recover 20% to Loyal
Never Purchased	≈28% Cus	0%	Welcome sequence 3-email + first-purchase discount	Activate 5% (≈1,700 new Cus)

Note: Cus = Customer

Impact estimation. If a win-back campaign successfully recovers 20% of the At Risk segment (approximately 2,050 customers with an average monetary value of $\approx 161,523$ VND), the business could recover around 331 million VND in potential revenue. At the same time, activating 5% of the Never Purchased segment ($\approx 1,700$ customers with an estimated first-order value of $\approx 50,000$ VND) would generate an additional 85 million VND in incremental revenue—bringing the total to ≈ 416 million VND from two low-cost interventions. (Data note: All analyses are conducted on a snapshot as of 31/12/2022.)

Summary of 5 key findings: First, concentration risk: 64.1% of revenue depends on a single segment (Champions) requires immediate protection and diversification. **Second, revenue at risk:** $\approx 13.8\%$ of revenue is under threat from At Risk + Lost + About To Sleep segments. **Next, onboarding gap:** $Month - 1$ retention is only 3.4%—96.6% of customers do not return after their first purchase. **Fourth potential loyalists paradox:** Highest AOV but low frequency focus should be on increasing purchase frequency, not order value. **Lastly, channel insight:** organic_search delivers both high volume and high quality, should be the priority channel to scale.

2 Approach, Modeling Pipeline, and Experimental Results

2.1 Approach

The task is to forecast `Revenue` and `COGS` on a daily basis (01/01/2023 to 01/07/2024) using the training set (04/07/2012 to 31/12/2022). The approach prioritizes robustness and prevention of data leakage, while remaining closely aligned with insights from EDA.

Feature Engineering (≈ 70 features). Features are constructed based on EDA findings to maximize the model's extrapolation capability.

Calendar & Events. Includes `dow`, `month`, `quarter`, 10 major Vietnamese holidays, and a Tet countdown variable of ± 14 days to capture fixed annual events.

Fourier Terms. Applies $\sin / \cos$ components with yearly/weekly/monthly cycles to accurately model strong seasonality in the time series.

Regime & Trend. Marks volatility periods such as `covid`, `post2021`, and 6 static promotional campaigns. A linear variable `t_days` is used to capture growth trends. `web_traffic` is excluded to prevent data leakage as it is not available in the test set.

Explainability (SHAP & Feature Importance). SHAP analysis confirms that `t_days` is the most important feature (*Top 1 Feature Importance*), driving the growth trend; Fourier terms capture Tet seasonality; `dow` learns weekend consumption patterns; and `regime_covid` explains the pandemic shock. These findings fully align with EDA hypotheses.

2.2 Model Architecture and Pipeline

The architecture combines the **Mixture of Experts (MoE)** concept with Hybrid Forecasting to handle both linear trends and seasonality. The pipeline consists of three stages:

Preprocessing & Augmentation. Target transformation $\tilde{y} = \log(Revenue)$. Data is augmented with Gaussian noise: $\tilde{y}^{(\pm)} = \tilde{y} \pm \mathcal{N}(0, 0.03\sigma_{\tilde{y}})$, tripling the dataset. Sample weights are applied: w=1.0 (2014–2018) and w=0.01 (pre-2014) to focus learning on recent trends.

M1 – Ridge Regression. Acts as a trend-capturing model, addressing the poor extrapolation capability of tree-based methods.

M2 – LightGBM Two-Stage. The nonlinear forecasting core. Uses **Time-Series Cross-Validation** (with the last 6 months of 2022 as a hold-out set) and early stopping to determine `best_iter`, preventing overfitting, followed by retraining on the full dataset.

M3 – Q – Specialists. Applies a "divide-and-conquer" strategy, training 8 specialized LGBM models (4 quarters $\times$ 2 targets). By doubling the weights of observations within the target quarter, these specialists capture fine-grained seasonal behaviors effectively.

Ensemble Architecture & Post-processing. Layer 1 blending: $\hat{y}_{blend} = 0.6\hat{y}_{Q-Spec} + 0.4\hat{y}_{LGB}$, prioritizing highly sensitive micro-patterns from Q-Specialists. Layer 2 blending: $\hat{y}_{raw} = 0.1\hat{y}_{Ridge} + 0.9\hat{y}_{blend}$, injecting Ridge's extrapolation strength to maintain trend continuity. Finally, apply Scale Calibration factors $CR \approx 1.26, CC \approx 1.32$ to adjust for strong post-COVID recovery momentum, along with a Margin Fix to preserve the macro-level COGS margin.

2.3 Experimental results

Table 1: Revenue forecasting performance on the internal validation set.(01/07/2022 – 31/12/2022)

Model	**MAE** $\downarrow$	**RMSE** $\downarrow$	**R²** $\uparrow$
M1 – Ridge Regression	543,799	773,416	0.6012
M2 – LightGBM Two-Stage	444,073	607,048	0.7543
M3 – Q-Specialists	$\approx$430,000	$\approx$590,000	$\approx$0.7650
Ensemble Raw (Before Calibration)	**$\approx$415,000**	**$\approx$575,000**	**$\approx$0.7750**